Linear Algebra Sample Exam 1D

Work done on scratch paper will not be graded. You don't need to show your work on pages 1 and 2.

1. (5 points) Indicate whether the situations are ***possible*** or ***impossible***.

		possible	impossible
a)	A is a 5×4 matrix with pivots in every column, $\vec{b} \in \mathbb{R}^5$ is not in the span of the columns of A.	◯	◯
b)	A is a 5×4 matrix that has 4 pivotal columns, and the columns of A are linearly dependent.	◯	◯
c)	Vector $\vec{x}$ is in $\mathbb{R}^n$, every row of the $m \times n$ matrix A contains a pivot, and the linear system $A\vec{x} = \vec{b}$ has an infinite number of solutions for some $\vec{b} \in \mathbb{R}^m$.	◯	◯
d)	$T(\vec{x}) = A\vec{x}$ is an onto linear transform and $A\vec{x} = \vec{0}$ has non-trivial solutions.	◯	◯
e)	$T(\vec{x}) = A\vec{x}$ is a linear transform, $\vec{x}$ is a vector in $\mathbb{R}^2$, A is a 2×2 matrix with exactly one pivot, and the only vector in the range of the transform is the zero vector.	◯	◯

2. (7 points) Indicate ***true*** if the statement is true, otherwise, indicate ***false***.

		true	false
a)	If $\vec{e}_1$ and $\vec{e}_2$ are the standard vectors in $\mathbb{R}^3$, and $T(\vec{x}) = A\vec{x}$ is a matrix transform that satisfies $T(\vec{e}_1) = T(\vec{e}_2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, then every entry of A is equal to 1.	◯	◯
b)	If A is $m \times n$, $\vec{y}_1$ and $\vec{y}_2$ are vectors in $\mathbb{R}^n$, $\vec{y}_1 \neq \vec{y}_2$, and $A\vec{y}_1 = A\vec{y}_2$, then $T_A(\vec{x}) = A\vec{x}$ cannot be onto.	◯	◯
c)	If $A\vec{x} = \vec{b}$ is consistent, then every column of A must be pivotal.	◯	◯
d)	If the only solution to the linear system $A\vec{x} = \vec{0}$ is the trivial solution, the columns of A must be linearly independent.	◯	◯
e)	Matrix A is 3×5, and $A\vec{x} = \vec{b}$ has a solution for all $\vec{b} \in \mathbb{R}^3$. The columns of A span $\mathbb{R}^3$.	◯	◯
f)	If the linear system $A\vec{x} = \vec{b}$ is consistent, then the augmented matrix $(A \mid \vec{b})$ does not have a pivot in the last column.	◯	◯
g)	Suppose A and E are $n \times n$ matrices, and $\vec{x} \in \mathbb{R}^n$. If $A\vec{x} = \vec{0}$ and A is row equivalent to E, then $E\vec{x} = \vec{0}$.	◯	◯